INBORN ERRORS OF NUCLEIC ACID METABOLISM

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Introduction

Purine Degradation

AMP → Adenosine → Inosine → Hypoxanthine → Xanthine → Uric acid

GMP → Guanosine → Guanine → Xanthine → Uric acid

Important enzyme: Xanthine oxidase

Hypoxanthine → Xanthine

Xanthine → Uric acid

Introduction

Pyrimidine Degradation

Pyrimidine degradation produces water-soluble products.

Uracil → β-alanine
Thymine → β-aminoisobutyrate

The products can enter other metabolic pathways.

Pyrimidine degradation does not produce uric acid.

INTRODUCTION

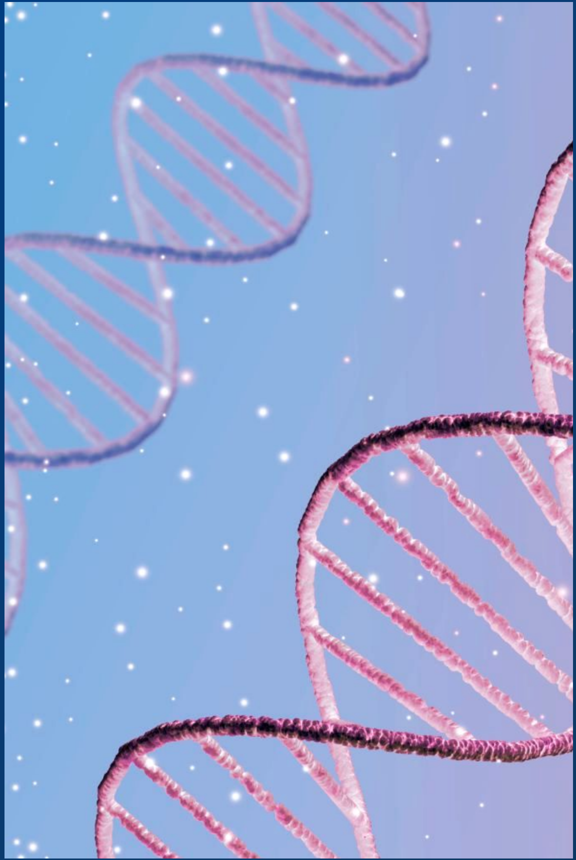
Inborn errors of metabolism are **inherited defects** in specific metabolic pathways.

Nucleic acid metabolism involves the synthesis and degradation of purines and pyrimidines.

Defects in these pathways may lead to **accumulation or deficiency** of specific metabolites.

Important disorders include:

1. Lesch-Nyhan syndrome
2. Gout
3. Adenosine deaminase deficiency
4. PNP deficiency
5. Orotic aciduria



GOUT:

Gout results from increased concentration of uric acid in blood **(hyperuricemia).**

Causes:

1. Increased production of uric acid (more common)
2. Decreased renal excretion of uric acid (less common)

HYPERURICEMIA

- Excess of uric acid in blood

FEMALE : 2.4 – 6.0 mg/dl

MALE : 3.4 – 7.0 mg/dl

TYPES :

1. Primary hyperuricemia
2. Secondary hyperuricemia

PRIMARY HYPERURICEMIA:

- Increased production of uric acid from purines
- Kidney cannot get rid of uric acid in blood
- Altered uric acid excretion can result from the decreased tubular secretion, increased glomerular filtration, or enhanced tubular absorption

SECONDARY HYPERURICEMIA:

- Certain cancer or chemotherapy
- Kidney diseases
- Medications
- Endocrine or metabolic conditions, etc.

Effect of hyperuricemia:

Deposition of monosodium urate crystal (MSU) in the joints

⇩

Inflammatory response of these crystals

⇩

First ACUTE then CHRONIC gouty ARTHISITIS

TOPHI:

Nodular masses of MSU crystals (tophi) may be deposited in the soft tissues

⇩

CHRONIC TOPACEOUS GOUT

UROLITHIASIS

Formation of uric acid stones in the kidney

Under excretion of uric acid :

Primary cause is unidentified inherent excretory defect

or

Secondary to any disease (lactic acidodsis), environmental factors (thiazide diuretic), lead exposure

Over production of uric acid:

Could be **idiopathic**

Or

Mutation in gene for X linked PRPP synthetase, leading to increase availability of PRPP hence increase production of purine .

TREATMENT:

PROBENECIDS OR SULFINPYRAZON

Increase renal excretion of uric acid

ALLOPURINOL

Inhibits **xanthine oxidase**

↓ conversion of hypoxanthine and xanthine to uric acid

Therefore ↓ uric acid production

ADENOSINE DEAMINASE DEFICIENCY(ADA)

Causes

ADA deficiency causes accumulation of:

- **Adenosine**
- **Deoxyadenosine**
- **dATP**

Increased dATP inhibits ribonucleotide reductase, impairing DNA synthesis

Untreated ADA Deficient children usually die before age of 2 from overwhelming infections

Result

Impaired lymphocyte development
→ **Severe combined immunodeficiency (SCID)**,involving T cells , B cells , Natural Killer Cells depletion **(lymphocytopenia)**

Treatment

- Bone marrow transplantation
- Enzyme replacement therapy
- Gene therapy

Lesch-Nyhan Syndrome

X linked recessive disorder

Cause: Deficiency of HGPRT (hypoxanthine-guanine phosphoribosyltransferase).

HGPRT is part of the purine salvage pathway.

RESULTS

Normally:

Hypoxanthine → IMP
Guanine → GMP

When HGPRT is deficient:

↓ Purine salvage
↑ PRPP
↑ de novo purine synthesis
↑ uric acid production

- uric acid stones in kidney (urolithiais)
- deposition of urate crystals in joints (gouty arthritis) and soft tissues
- motor dysfunctions
- cognitive defects
- behavioral disturbances
- self mutilations includes
- biting of lips and fingers

PURINE NUCLEOSIDE PHOSPHORYLASE (PNP) DEFICIENCY

- Autosomal recessive deficiency
- Rarer and less severe than ADA deficiency
- Affects **Tcells developmemt primarily**
- PNP deficient individuals have **recurrent infections and neurodevelopmental delay**

Orotic Aciduria

Cause: Deficiency of enzymes involved in de novo pyrimidine synthesis:

UMP synthase deficiency

This enzyme has two activities:

1. Orotate phosphoribosyltransferase
2. Orotidine-5'-phosphate decarboxylase

Defect causes:

→ ↑ Orotic acid

→ ↓ UMP

→ Impaired pyrimidine synthesis

TREATMENT:

Administration of URIDINE results in impprovement of anemia and decreased excreation of orotate

Features include:

- Megaloblastic anemia
- Growth retardation
- Developmental problems
- Increased urinary orotic acid

SUMMARY

Lesch-Nyhan Syndrome

HGPRT deficiency
↑ Uric acid
Neurological abnormalities

Usually overproduction/underexcretion
↑ Uric acid
Urate crystals

Gout

ADA deficiency

Adenosine deaminase deficiency
↑ dATP SCID

UMP synthase deficiency
↑ Orotic acid
Megaloblastic anemia

Orotic aciduria

THANK YOU

LIPPINCOTT EIGHTH EDITION